A REPRESENTABLE MORPHISM IS A UNIVERSAL HOMEOMORPHISM EXACTLY WHEN ALL ITS BASE CHANGES ARE HOMEOMORPHISMS ON SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the Stacks Project keeps one single shared preamble for every chapter; this card takes it directly from the staged sources/stacks-algebraic-stacks-morphisms/preamble.tex as three verbatim spans (lines 8-12, 14-18 and 154-217), so no derived file was written and nothing was added or redefined. The wrapper class and the base packages are supplied by the assembler.

Editorial note. Bibliography provenance: neither the counted statement nor its proof contains a citation (this lane rescans the whole span before writing the card), so the card carries no bibliography and no entry was reconstructed.

Editorial note. Selection provenance: this card was opened under the file-based control contract (CONTROL.md), with the mechanical pre-card gate run on its statement and proof interval before the card was written.

Allowed standard prerequisites (editorial).

- The definition of a universal homeomorphism for representable morphisms of algebraic stacks, tested after base change by schemes. Referred to in the statement (source lines 3105-3106) and in the proof for the second implication. Formulates the two conditions being compared.
- The description of the topological space of an algebraic stack through a smooth cover, with the induced maps open and surjective and the fibre product square surjective onto the fibre product of the spaces. Invoked inside the counted proof (source lines 3119-3134) as a pointer to the chapter on properties of stacks. Descends the homeomorphism property from a smooth cover to the base change of the stack.

Source reference. Original reference stacks-properties-section-points: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 2488–2488. Navigation pointer only: the counted text refers to the Stacks item stacks-properties-section-points, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference stacks-properties-section-properties-morphisms: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 248–248. Navigation pointer only: the counted text refers to the Stacks item stacks-properties-section-properties-morphisms, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its

Date: stacks-project:stacks-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks which is representable by algebraic spaces. The following are equivalent*

(1) *f is a universal homeomorphism (Properties of Stacks, Section stacks-properties-section-properties-morphisms), and*
(2) *for every morphism of algebraic stacks $\mathcal{Z} \to \mathcal{Y}$ the map of topological spaces $|\mathcal{Z} \times_{\mathcal{Y}} \mathcal{X}| \to |\mathcal{Z}|$ is a homeomorphism.*

Proof. Assume (1), and let $\mathcal{Z} \to \mathcal{Y}$ be as in (2). Choose a scheme V and a surjective smooth morphism $V \to \mathcal{Z}$. By assumption the morphism $V \times_{\mathcal{Y}} \mathcal{X} \to V$ of algebraic spaces is a universal homeomorphism, in particular the map $|V \times_{\mathcal{Y}} \mathcal{X}| \to |V|$ is a homeomorphism. By Properties of Stacks, Section stacks-properties-section-points in the commutative diagram

$$\begin{array}{ccc} |V \times_{\mathcal{Y}} \mathcal{X}| & \longrightarrow & |\mathcal{Z} \times_{\mathcal{Y}} \mathcal{X}| \\ \downarrow & & \downarrow \\ |V| & \longrightarrow & |\mathcal{Z}| \end{array}$$

the horizontal arrows are open and surjective, and moreover

$$|V \times_{\mathcal{Y}} \mathcal{X}| \longrightarrow |V| \times_{|\mathcal{Z}|} |\mathcal{Z} \times_{\mathcal{Y}} \mathcal{X}|$$

is surjective. Hence as the left vertical arrow is a homeomorphism it follows that the right vertical arrow is a homeomorphism. This proves (2). The implication (2) $\Rightarrow$ (1) follows from the definitions. □